

A Few Series

1. Does $\sum_{k=0}^{\infty} \frac{2^k}{3^k}$ converge? If so, what is the limit? *yes, $\frac{1}{1-\frac{2}{3}} = 3$*

2. Does $\left\{ \frac{2^k}{3^k} \right\}$ converge? If so, what is the limit? *yes, $\lim_{k \rightarrow \infty} \frac{2^k}{3^k} = 0$ (Some will think it's $\frac{2}{3}$)*

Do the following series converge or diverge? State the test you used. For convergent series, indicate if they converge conditionally or absolutely. If appropriate, find the sum.

3. $\sum_{k=1}^{\infty} \frac{k-1}{k2^k}$ *conv. ratio test or root test or limit comp w/ $\sum \frac{1}{2^k}$*

4. $\sum_{k=1}^{\infty} \frac{7^k}{k^2}$ *Div. ratio or root or Divergence Test*

5. $\sum_{k=1}^{\infty} (-1)^k \frac{k^2}{k!}$ *conv. abs. Alt. Series Test first, then ratio test or $\sum \frac{k^2}{k!}$*

6. $\sum_{k=1}^{\infty} ke^{-k}$ *conv. Ratio or root or even Integral Test*

7. $\sum_{k=2}^{\infty} \frac{1}{k(\ln k)^2}$ *conv. Int. Test ($\int \frac{1}{u^2} du$)*

8. $\sum_{k=1}^{\infty} \frac{\sin k}{k\sqrt{k}}$ *conv. absolutely $\left| \frac{\sin k}{k\sqrt{k}} \right| \leq \frac{1}{k^{3/2}}$ Direct Comp. Test*

9. $\sum_{k=2}^{\infty} \frac{1 + \ln k}{k^3}$ *conv. Limit Comp w/ log-g $\sum \frac{\ln k}{k^3}$ or Zero End. Test w/ $\sum \frac{1}{k^2}$*

10. $\sum_{k=1}^{\infty} \frac{\cos k\pi}{k+3}$ *conv. conditionally $\cos k\pi = (-1)^k$, Alt. Series Test, then LCT w/ $\sum \frac{1}{k}$*

11. $\sum_{k=2}^{\infty} \frac{3^{k-1} 2^{k+2}}{7^{k+1}}$ *conv. ratio or root or re-write as $\sum 3 \left(\frac{6}{7}\right)^k$*

12. $\sum_{k=1}^{\infty} \frac{\ln k}{\sqrt{3k+2}}$ *~~conv.~~ Div. Direct Comp to $\sum \frac{1}{\sqrt{3} \sqrt{k}}$ or LCT w/ $\sum \frac{\ln k}{\sqrt{k}}$ etc*

13. Find all values of x for which the series $\sum_{k=3}^{\infty} \frac{k(x-5)^k}{3^k}$ converges. (absolutely)

Important ~~TR~~ tell them to take abs value, then do ratio or root test.

$$\lim_{k \rightarrow \infty} \frac{(k+1) |x-5|^{k+1}}{3^{k+1}} \cdot \frac{3^k}{k |x-5|^k} = \frac{|x-5|}{3} < 1 \Rightarrow |x-5| < 3$$

$\Rightarrow x > \text{dist from } x \text{ to } 5 < 3$
so x is between 2 and 8

check $x=8 \leq \frac{k \cdot 3^k}{3^k}$ div.

check $x=2 \leq k(-1)^k$ div

$(2, 8)$

Worksheet

Do the following series converge or diverge? State the test you used. For convergent series, indicate if they converge conditionally or absolutely. If appropriate, find the sum.

1. $\sum_{k=1}^{\infty} \frac{\cos k}{k^2}$ *Conv.* $\left| \frac{\cos k}{k^2} \right| < \frac{1}{k^2}$

2. $\sum_{k=1}^{\infty} \frac{k-1}{k^{3/2}}$ ~~Div.~~ *Div.* *LCT w/ $\sum \frac{1}{\sqrt{k}}$*

3. $\sum_{k=1}^{\infty} k \sin\left(\frac{1}{k}\right)$ *Tricky one* $\lim_{k \rightarrow \infty} \frac{\sin\left(\frac{1}{k}\right)}{\frac{1}{k}} = \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

So Div. by LCT, compare w/ $\sum \frac{1}{k}$

4. $\sum_{k=1}^{\infty} \frac{(-1)^k 7^{2k}}{(2k)!}$ *Conv. absolutely* *Take abs, then Ratio Test*

5. Can the ratio test be used for the series $\sum_{k=0}^{\infty} \frac{(-1)^k k^2}{3^k}$? Completely explain your answer.

Technically, no. Ratio Test in text requires positive terms.

Take $\sum \left| \frac{(-1)^k k^2}{3^k} \right|$, then test for absolute convergence.